

Date: August 15, 2025

## Algorithms -1-

University of Aleppo

Student Name:

Model: C

Duration: 90 Minutes

Student Number:

Select the correct answer for each question:

1. Find the area of a rectangle of side  $a = 6$   
A. 34                      B. 36                      C. 40                      D. 31                      [1 points]
2. Solve for  $x$  in the equation  $2x + 6 = 0$   
A. -1                      B. -2                      C. -3                      D. 3                      [1 points]
3. Calculate the derivative of  $f(x) = x^2$  then find the limit to infinity of the function  $\lim_{x \rightarrow \infty} f(x)$   
A. 14                      B. 15                      C. 16                      D. 17                      E. 17                      [1 points]
4. Test  
A. 13                      B. 14                      C. 15                      D. 16                      E. 17                      [3 points]
5. For any positive integer  $x$  what is the value of  $\int \int \int x^5$ ?  
A. 30                      B. 40                      C. 50                      D. 60                      [1 points]
6. Hello!  
A. 1                      B. 2                      C. 3                      D. 4                      [2 points]
7.  $x = 3 + 6$  what is the value of  $x$ ?  
A. 7                      B. 9                      C. 8                      D. 10                      [1 points]
8. What is the derivative of  $x^2$   
A.  $x$                       B.  $2 * x$                       [5 points]
9. Calculate  $\sum i$   
A.  $n * (n + 1) / 3$                       B.  $n * (n - 1) / 2$                       [1 points]
10. Please calculate  $\int f(x) dx$   
A. 13                      B. 14                      C. 15                      D. 16                      [2 points]